

UDC - Universal Data Converter

Demonstration prototype for organizing data conversion workflows with Python and Microsoft SQL Server.

Purpose	Standardized presentation, inspection, import validation and export of database records.
Technology	Python / Flask, Microsoft SQL Server, Docker, GitHub Codespaces.
Current export	CSV, JSON and XML.
Current import	CSV validation workflow; database write integration is planned for a later stage.
Data	All data shown in the prototype is synthetic and intended only for testing and demonstration.

Project status: Early demonstration prototype, not a production-ready system. Further development includes full CSV-to-database import integration, extended field mapping, stronger validation, improved error handling and additional automated tests.

Live demo access is available in two ways. The primary link connects via a secure Cloudflare Tunnel directly to the application running on the author's PC and displays only synthetic demo data from this local test environment. It remains available while the PC and tunnel are running and is not subject to GitHub Codespaces idle-time limits. An alternative instance runs in the free GitHub Codespaces environment, which may stop automatically after periods of inactivity and can be restarted on request. Both versions can be opened directly in a browser; no local installation is required.

Application overview

Home screen showing the database viewer, export and import modules, together with the SQL Server connection status.

Converter_2 / UDC [Home](#) [DB View](#) [Export](#) [Import](#) [About](#)

Universal Data Converter

SQL Server database viewer, import and export interface.

Database View

Browse SQL Server tables and views.

[Open DB View](#)

Export

Export standardized UDC data to CSV, JSON or XML.

[Open Export](#)

Import

Upload CSV, JSON or XML data for UDC import.

[Open Import](#)

System

Application	Converter_2 / UDC
Author	LiR
Server	mssql_2025_dev,1433
Database	Converter_UDC
Connection	Connected

Database View

Browser-based inspection of SQL Server tables and views using synthetic demonstration data.

Converter_2 / UDC Home **DB View** Export Import About

Database View

Table / View

Rows

dbo.data_measurement ▾ 100 ▾ **Load**

Object: **dbo.data_measurement** · rows returned: **9**

ID	PatientID	PatientName	MeasurDate	ParameterID	ParameterName	MeasurVal	UnitCode	SourceSystemID	SourceSystemName	ImportBa
1	1	patient1	2026-09-17 09:00:00	1	Glucose	5.800000	mmol/L	1	TestSource	None
2	1	patient1	2026-09-17 09:05:00	2	Temperature	36.700000	C	1	TestSource	None
3	1	patient1	2026-09-17 09:10:00	3	CRP	2.400000	mg/L	1	TestSource	None
4	2	patient2	2026-09-17 10:00:00	1	Glucose	6.200000	mmol/L	1	TestSource	None
5	2	patient2	2026-09-17 10:05:00	2	Temperature	37.200000	C	1	TestSource	None
6	2	patient2	2026-09-17 10:10:00	3	CRP	6.800000	mg/L	1	TestSource	None
7	3	patient3	2026-09-17 11:00:00	1	Glucose	4.900000	mmol/L	1	TestSource	None
8	3	patient3	2026-09-17 11:05:00	2	Temperature	38.100000	C	1	TestSource	None
9	3	patient3	2026-09-17 11:10:00	3	CRP	18.500000	mg/L	1	TestSource	None

Export

Export workflow for standardized UDC data. The current prototype supports CSV, JSON and XML.

Converter_2 / UDC

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Export Data

Test implementation: Export supports CSV, JSON and XML.

1. Export format

CSV

2. Destination folder:

app/export

Choose another folder

3. Generate & Save

Every export is created in app/export. Choosing another folder additionally saves a local copy there.

Ready export examples

These files are already included in **app/export** and show the result of the implemented export formats.

File	Format	Size
ExportUDC_23_09_2026_12_31_54.csv	CSV	1037 bytes
ExportUDC_23_09_2026_09_15_09.xml	XML	3995 bytes
ExportUDC_23_09_2026_09_15_09.json	JSON	3343 bytes
ExportUDC_23_09_2026_09_15_09.csv	CSV	1037 bytes
ExportUDC_18_09_2026_16_22_14.csv	CSV	1025 bytes
ExportUDC_18_09_2026_16_22_14.xml	XML	3995 bytes
ExportUDC_18_09_2026_16_22_14.json	JSON	3343 bytes

Recent operations

Time	Operation	File	Status	Details
12:44:06	EXPORT	ExportUDC_23_09_2026_12_31_54.csv	OK	1037 bytes · app/export

Import

CSV import validation with ready-to-use test files, row validation and recent operation status.

Converter_2 / UDC

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Import Data

Test implementation: Import currently supports CSV only.

Ready-to-use test files

These CSV files are already included in **app/import**. For the test task, click **Load test file** — no file search is required.

Test file	Location	Action
ClientImport_A.csv	app/import	<button>Load test file</button>
ClientImport_B.csv	app/import	<button>Load test file</button>
ClientImport_C.csv	app/import	<button>Load test file</button>

The current web test validates the CSV structure and rows. The database import dispatcher will be connected separately.

Upload your own CSV

CSV source file

Choose File

No file chosen

Upload CSV

Import test completed

File

ClientImport_A.csv

Rows validated

3

Target

dbo.data_measurement

Stage

CSV validation completed

Prototype note: the current web flow validates the CSV. Database write is not connected yet.

Recent operations

Time	Operation	File	Status	Details
12:44:46	IMPORT	ClientImport_A.csv	OK	3 rows validated · ready test file

Project links

GitHub repository / README:

<https://github.com/RLisovenko/UDC-UniversalDataConverter>

Direct live demo - UDC_UniversalDataConverter_Demo (author's PC):

<https://sonic-ace-dragon-threaded.trycloudflare.com>

This link connects through a secure Cloudflare Tunnel directly to the application running on the author's PC. It uses only synthetic data from the author's local test environment and is available while the PC and tunnel are running. It is not subject to GitHub Codespaces idle-time limits.

Alternative live demo - GitHub Codespaces (free environment):

<https://ominous-sniffle-jrr7qpg9pv9cjqqj-5000.app.github.dev/>

The free Codespaces environment may stop automatically after periods of inactivity. If it is offline, it can be restarted on request. You may try either live-demo link in a browser; no local installation is required.

Data / safety note:

Both demonstrations use only synthetic test data and are completely separate from Klinikum Oldenburg systems and databases. No Klinikum data is read or written.

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